DATA SAMPLING AND SAMPLING DISTRIBUTIONS

Objectives

- Differentiate sample and population.
- Discuss the types of sampling methods.
- Define sampling distribution and its applications.

Introduction

- A popular misconception holds that the era of big data means the end of a need for sampling.
- In fact, the proliferation of data of varying quality and relevance reinforces the need for sampling as a tool to work efficiently with a variety of data and to minimize bias.
- Even in a big data project, predictive models are typically developed and piloted with samples.
- Samples are also used in tests of various sorts.

Random Sampling and Sample Bias

- A **sample** is a subset of data from a larger data set; statisticians call this larger data set the population.
- A **population** in statistics is not the same thing as in biology—it is a large, defined (but sometimes theoretical or imaginary) set of data.

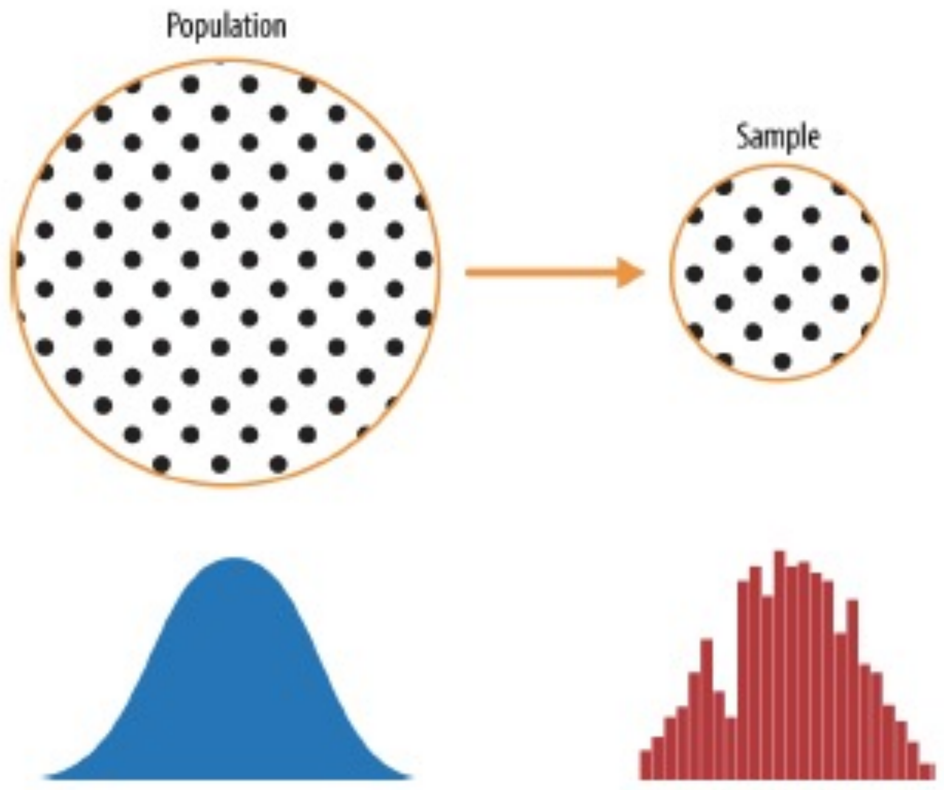


Figure 2-1. Population versus sample

Random Sampling and Sample Bias

- *Random sampling* is a process in which each available member of the population being sampled has an equal chance of being chosen for the sample at each draw.
- The sample that results is called a *simple random sample*.
- Sampling can be done with replacement, in which observations are put back in the population after each draw for possible future reselection.
- Or it can be done without replacement, in which case observations, once selected, are unavailable for future draws.

The importance of Data Quality

- Data quality often matters more than data quantity when making an estimate or a model based on a sample.
- Data quality in data science involves completeness, consistency of format, cleanliness, and accuracy of individual data points.
- Statistics adds the notion of representativeness.

Key Terms for Random Sampling

Sample
: A subset from a larger data set.

Population
: The larger data set or idea of a data set.

N (n)
: The size of the population (sample).

Random sampling
: Drawing elements into a sample at random.

Stratified sampling
: Dividing the population into strata and randomly sampling from each strata.

Stratum (pl., strata)
: A homogeneous subgroup of a population with common characteristics.

Simple random sample
: The sample that results from random sampling without stratifying the population.

Bias
: Systematic error.

Sample bias
: A sample that misrepresents the population.

The 1936 Poll: Alf Landon vs. Franklin Roosevelt

The 1936 Poll: Alf Landon vs. Franklin Roosevelt

- The classic example is the Literary Digest poll of 1936 that predicted a victory of Alf Landon over Franklin Roosevelt.
- The Literary Digest, a leading periodical of the day, polled its entire subscriber base plus additional lists of individuals, a total of over 10 million people, and predicted a landslide victory for Landon.
- George Gallup, founder of the Gallup Poll, conducted biweekly polls of just 2,000 people and accurately predicted a Roosevelt victory.
- *The difference lay in the selection of those polled.*

Bias

- ***Statistical bias*** refers to measurement or sampling errors that are systematic and produced by the measurement or sampling process.
- An important distinction should be made between errors due to random chance and errors due to bias.

Bias

- Bias comes in different forms and may be observable or invisible.
- When a result does suggest bias (e.g., by reference to a benchmark or actual values), it is often an indicator that a statistical or machine learning model has been misspecified, or an important variable left out.

Random Selection

- To avoid the problem of sample bias that led the Literary Digest to predict Landon over Roosevelt, George Gallup opted for more scientifically chosen methods to achieve a sample that was representative of the US voting electorate.
- There are now a variety of methods to achieve representativeness, but at the heart of all of them lies ***random sampling***.

George Gallup, catapulted to fame by the Literary Digest's "big data" failure

Inferential Statistics

- Goal: Make inferences about the population based on data from a sample.
- Inference may be defined as the process of drawing conclusions based on evidence and reasoning.

Five (5) Common Sampling Methods

- Simple Random Sampling
- Stratified Random Sampling
- Cluster Sampling
- Systematic 1-in-k Sampling
- Convenience Sampling

Sampling Plan

- The way a sample is selected is called the *sampling plan* or experimental design and determines the quantity of information in the sample.

Simple Random Sampling

- Simple random sampling is a commonly used sampling plan in which every sample of size n has the same or equal chance of being selected.
- For example, suppose you want to select a sample of size $n=2$ from a population containing $N=4$ objects.

Simple Random Sampling

- If the sample of n = 2 observations is selected so that each of these six samples has the same chance of selection, given by 1/6, then the resulting sample is called a simple random sample, or just a random sample.

Ways of Selecting a Sample of Size 2 from 4 Objects

Sample	Observations in Sample
1	x_1, x_2
2	x_1, x_3
3	x_1, x_4
4	x_2, x_3
5	x_2, x_4
6	x_3, x_4

Simple Random Sampling

- Select a random sample of size 5 from this class.

Definition If a sample of n elements is selected from a population of N elements using a sampling plan in which each of the possible samples has the same chance of selection, then the sampling is said to be **random** and the resulting sample is a **simple random sample.**

FORMAL DEFINITION

Example

- A computer database at a downtown law firm contains files for N=1000 clients. The firm wants to select n=5 files for review. Select a simple random sample of 5 files from this database.

Solution

- You must first label each file with a number from 1 to 1000.
- Perhaps the files are stored alphabetically, and the computer has already assigned a number to each.
- Then generate a sequence of 10 three-digit random numbers.
- The random starting point ensures that you will not use the same sequence over and over again.

Solution

Portion of a Table of Random Numbers

15574	35026	98924
45045	36933	28630
03225	78812	50856
88292	26053	21121

TABLE 10 Random Numbers

Line	Column 1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	10480	15011	01536	02011	81647	91646	69179	14194	62590	36207	20969	99570	91291	90700
2	22368	46573	25595	85393	30995	89198	27982	53402	93965	34095	52666	19174	39615	99505
3	24130	48360	22527	97265	76393	64809	15179	24830	49340	32081	30680	19655	63348	58629
4	42167	93093	06243	61680	07856	16376	39440	53537	71341	57004	00849	74917	97758	16379
5	37570	39975	81837	16656	06121	91782	60468	81305	49684	60672	14110	06927	01263	54613
6	77921	06907	11008	42751	27756	53498	18602	70659	90655	15053	21916	81825	44394	42880
7	99562	72905	56420	69994	98872	31016	71194	18738	44013	48840	63213	21069	10634	12952
8	96301	91977	05463	07972	18876	20922	94595	56869	69014	60045	18425	84903	42508	32307
9	89579	14342	63661	10281	17453	18103	57740	84378	25331	12566	58678	44947	05585	56941
10	84575	36857	53342	53988	53060	59533	38867	62300	08158	17983	16439	11458	18593	64952
11	28918	69578	88231	33276	70997	79936	56865	05859	90106	31595	01547	85590	91610	78188
12	63553	40961	48235	03427	49626	69445	18663	72695	52180	20847	12234	90511	33703	90322
13	09429	93969	52636	92737	88974	33488	36320	17617	30015	08272	84115	27156	30613	74952
14	10365	61129	87529	85689	48237	52267	67689	93394	01511	26358	85104	20285	29975	89868
15	07119	97336	71048	08178	77233	13916	47564	81056	97735	85977	29372	74461	28551	90707
16	51085	12765	51821	51259	77452	16308	60756	92144	49442	53900	70960	63990	75601	40719
17	02368	21382	52404	60268	89368	19885	55322	44819	01188	65255	64835	44919	05944	55157
18	01011	54092	33362	94904	31273	04146	18594	29852	71585	85030	51132	01915	92747	64951
19	52162	53916	46369	58586	23216	14513	83149	98736	23495	64350	94738	17752	35156	35749
20	07056	97628	33787	09998	42698	06691	76988	13602	51851	46104	88916	19509	25625	58104
21	48663	91245	85828	14346	09172	30168	90229	04734	59193	22178	30421	61666	99904	32812
22	54164	58492	22421	74103	47070	25306	76468	26384	58151	06646	21524	15227	96909	44592
23	32639	32363	05597	24200	13363	38005	94342	28728	35806	06912	17012	64161	18296	22851
24	29334	27001	87637	87308	58731	00256	45834	15398	46557	41135	10367	07684	36188	18510
25	02488	33062	28834	07351	19731	92420	60952	61280	50001	67658	32586	86679	50720	94953
26	81525	72295	04839	96423	24878	82651	66566	14778	76797	14780	13300	87074	79666	95725
27	29676	20591	68086	26432	46901	20849	89768	81536	86645	12659	92259	57102	80428	25280
28	00742	57392	39064	66432	84673	40027	32832	61362	98947	96067	64760	64585	96096	98253
29	05366	04213	25669	26422	44407	44048	37937	63904	45766	66134	75470	66520	34693	90449
30	91921	26418	64117	94305	26766	25940	39972	22209	71500	64568	91402	42416	07844	69618
31	00582	04711	87917	77341	42206	35126	74087	99547	81817	42607	43808	76655	62028	76630
32	00725	69884	62797	56170	86324	88072	76222	36086	84637	93161	76038	65855	77919	88006
33	69011	65795	95876	55293	18988	27354	26575	08625	40801	59920	29841	80150	12777	48501
34	25976	57948	29888	88604	67917	48708	18912	82271	65424	69774	33611	54262	85963	03547
35	09763	83473	73577	12908	30883	18317	28290	35797	05998	41688	34952	37888	38917	88050
36	91567	42595	27958	30134	04024	86385	29880	99730	55536	84855	29080	09250	79656	73211
37	17955	56349	90999	49127	20044	59931	06115	20542	18059	02008	73708	83517	36103	42791
38	46503	18584	18845	49618	02304	51038	20655	58727	28168	15475	56942	53389	20562	87338
39	92157	89634	94824	78171	84610	82834	09922	25417	44137	48413	25555	21246	35509	20468
40	14577	62765	35605	81263	39667	47358	56873	56307	61607	49518	89656	20103	77490	18062
41	98427	07523	33362	64270	01638	92477	66969	98420	04880	45585	46565	04102	46880	45709
42	34914	63976	88720	82765	34476	17032	87589	40836	32427	70002	70663	88863	77775	69348
43	70060	28277	39475	46473	23219	53416	94970	25832	69975	94884	19661	72828	00102	66794
44	53976	54914	06990	67245	68350	82948	11398	42878	80287	88267	47363	46634	06541	97809
45	76072	29515	40980	07391	58745	25774	22987	80059	39911	96189	41151	14222	60697	59583
46	90725	52210	83974	29992	65831	38857	50490	83765	55657	14361	31720	57375	56228	41546
47	64364	67412	33339	31926	14883	24413	59744	92351	97473	89286	35931	04110	23726	51900
48	08962	00358	31662	25388	61642	34072	81249	35648	56891	69352	48373	45578	78547	81788
49	95012	68379	93526	70765	10592	04542	76463	54328	02349	17247	28865	14777	62730	92277
50	15664	10493	20492	38391	91132	21999	59516	81652	27195	48223	46751	22923	32261	85653

Source: From *Handbook of Tables for Probability and Statistics*, 2nd ed., edited by William H. Beyer (CRC Press). Used by permission of William H. Beyer.

Observational Study

- The situation described earlier is called an observational study because the data already existed before you decided to observe or describe their characteristics.
- Most sample surveys, in which information is gathered with a questionnaire, fall into this category.

Frequently occurring problems when conducting a sample survey:

- Nonresponse: You have carefully selected your random sample and sent out your questionnaires, but only 50% of those surveyed return their questionnaires.
- Are the responses you received still representative of the entire population, or are they biased because only those people who were particularly opinionated about the subject chose to respond?

Frequently occurring problems when conducting a sample survey:

- Undercoverage: You have selected your random sample using telephone records as a database.
- Does the database you used systematically exclude certain segments of the population—perhaps those who do not have telephones?

Frequently occurring problems when conducting a sample survey:

- Wording bias: Your questionnaire may have questions that are too complicated or tend to confuse the reader.
- Possibly the questions are sensitive in nature—for example, “Have you ever used drugs?” or “Have you ever cheated on your income tax?”—and the respondents will not answer truthfully.

Frequently occurring problems when conducting a sample survey:

- Methods have been devised to solve some of these problems, but only if you know that they exist.
- If your survey is biased by any of these problems, then your conclusions will not be very reliable, even though you did select a random sample!

Stratified Random Sample

- When the population consists of two or more subpopulations, called strata, a sampling plan that ensures that each subpopulation is represented in the sample is called a stratified random sample.

Examples

- Stratify the class by gender, randomly selected 3 female and 3 male students
- Voter poll – stratify the nation by region, randomly selected 100 voters in each region

Formal Definition

Definition **Stratified random sampling** involves selecting a simple random sample from each of a given number of subpopulations, or **strata.**

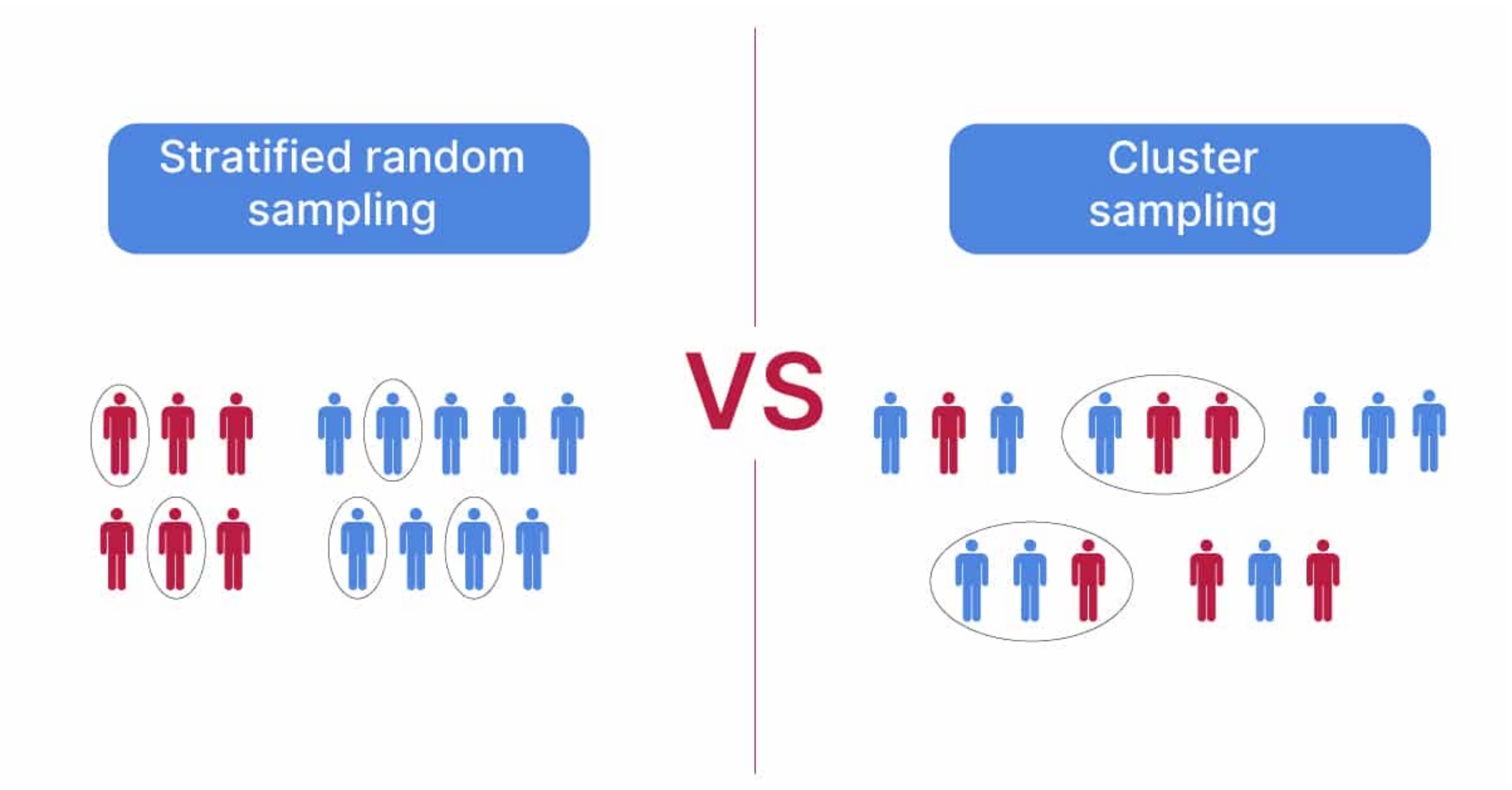
Cluster Sample

- Another form of random sampling is used when the available sampling units are groups of elements, called clusters.
- For example, a household is a cluster of individuals living together.
- A city block or a neighborhood might be a convenient sampling unit and might be considered a cluster for a given sampling plan.
- Students in an elementary school are clustered into classes.

Formal Definition

Definition A **cluster sample** is a simple random sample of clusters from the available clusters in the population.

Difference between Stratified and Random Sampling



Difference between Stratified and Random Sampling

- In Cluster Sampling, the sampling is done on a population of clusters therefore, cluster/group is considered a sampling unit.
- In Stratified Sampling, elements within each stratum are sampled.
- In Cluster Sampling, only selected clusters are sampled.
- In Stratified Sampling, from each stratum, a random sample is selected.
- In Cluster Sampling, the aim is to reduce cost and increase the efficiency of sampling.
- In Stratified Sampling, the motive is to increase precision to reduce error.

1-in-k systematic random sample

- Easiest way to draw a sample
- Obtain an ordered list of the population
- Select a random start point from 1,2,…,k
- Select every kth item in the list after your start point

Caution: sample could be biased

1-in-k systematic random sample

- Formal definition

Definition A **1-in-*k* systematic random sample** involves the random selection of one of the first *k* elements in an ordered population, and then the systematic selection of every *k*th element thereafter.

Convenience Sample

- A sample that is simple or easy to obtain without randomization.
- Examples:

–Internet polls

– Shopping mall polls

– A poll of your 5 closest friends

Convenience Sample

- These samples are often biased
- You can’t make inferences about the population from them

Selection bias

- *Selection bias* refers to the practice of selectively choosing data—consciously or unconsciously—in a way that leads to a conclusion that is misleading or ephemeral.
- *Selection bias* is a distortion in a measure of association (such as a risk ratio) due to a sample selection that does not accurately reflect the target population.
- Selection bias can occur when investigators use improper procedures for selecting a sample population.

Selection Bias

- Typical forms of selection bias in statistics, in addition to the vast search effect, include nonrandom sampling, cherry-picking data, selection of time intervals that accentuate a particular statistical effect, and stopping an experiment when the results look “interesting.”

Regression to the Mean

- *Regression to the mean* refers to a phenomenon involving successive measurements on a given variable: extreme observations tend to be followed by more central ones.
- Attaching special focus and meaning to the extreme value can lead to a form of selection bias.

Activity:

- A random sample of public opinion in a small town was obtained by selecting every 10th person who passed by the busiest corner in the downtown area.
- Will this sample have the characteristics of a random sample selected from the town's citizens? Explain.